PDPM IIITDM JABALPUR - CS3010 FUSION ERP LAB

Assignment 8: Specification-Driven AI-Assisted Testing Report

Assigned Module:	Other Academic Procedures (OAP) - Module 04
Subteam & Pair:	Subteam 04B | Rahul Kumar (24BCS205), Aakriti Karna (24BCS294)
A8_BASE Commit Hash:	4e4cb54 (Frozen Assignment 7 PR #505 baseline)
Assignment 8 Official PR:	https://github.com/FusionIIIT/Fusion-Integrated/pull/561
AI Model & Setup:	Qwen2.5-Coder 1.5B (qwen2.5-coder:1.5b via local Ollama server)
Testing Strategies:	FR (Functional Reqs) -> UC (Use Cases) -> BR (Business Rules) -> WF (Workflows)
Prescribed Framework:	pytest-django, pytest-cov, isolated SQLite in-memory test DB

1. Study Overview & Integrity Controls

Assignment 8 systematically verifies the frozen A8_BASE implementation of the OAP module across 4 independent specification perspectives: Functional Requirements (CFR/AFR), Use Cases (UC), Business Rules (BR), and Workflows (WF). All test designs (Prompt A) were frozen into JSON before implementation exposure (Prompt B). Production code in modules/oap was kept strictly unchanged, and identical local AI model controls were maintained across all four strategies.

2. Strategy Execution & Verification Results

Strategy	Spec Input	Obligations	Total	Passed	Failed	Skipped	Stmt Cov	Branch Cov
FR	Problem Statement + CFR/AFR	33 / 33	33	29	4	0	72%	0%
UC	Problem Statement + Use Cases	53 / 53	53	2	11	40	72%	0%
BR	Problem Statement + Business Rules	27 / 27	27	27	0	0	36%	0%
WF	Problem Statement + Workflows	11 / 11	11	11	0	0	36%	0%

3. Confirmed Implementation Defects (Defect Log Summary)

ISS-OAP-001 (Missing Functionality): StudentLeaveViewSet.hod_approve action calls non-existent hod_approve() on StudentLeaveRequest model, failing with 401/AttributeError when executed.

ISS-OAP-002 (Missing Functionality): BonafideViewSet.approve_and_issue action calls non-existent approve_and_issue() on BonafideRequest model, failing with 401/AttributeError when executed.

ISS-OAP-003 (Test-Generation Problem): Generated UC test cases queried unversioned path /api/oap/leave/ instead of /api/v1/oap/leave/, producing 404 Not Found.

4. Submission Deliverables & Artifact Index

All required study artifacts are preserved in the submission package Assignment_8_Submission_Subteam_04B.zip:

- Pair Excel Workbook: Assignment_8_Data_Collection_Subteam_04B.xlsx (All 4 sheets complete)
- Frozen Design Specifications: a8_design/{FR_DESIGN, UC_DESIGN, BR_DESIGN, WF_DESIGN}.json
- Automated Pytest Suites: tests/a8/{fr/test_fr.py, uc/test_uc.py, br/test_br.py, wf/test_wf.py}
- Machine Evidence: a8_results/{fr, uc, br, wf}/{junit.xml, coverage.xml, pytest.log}
- Production Integrity: modules/oap unchanged; 100% adherence to study protocols.